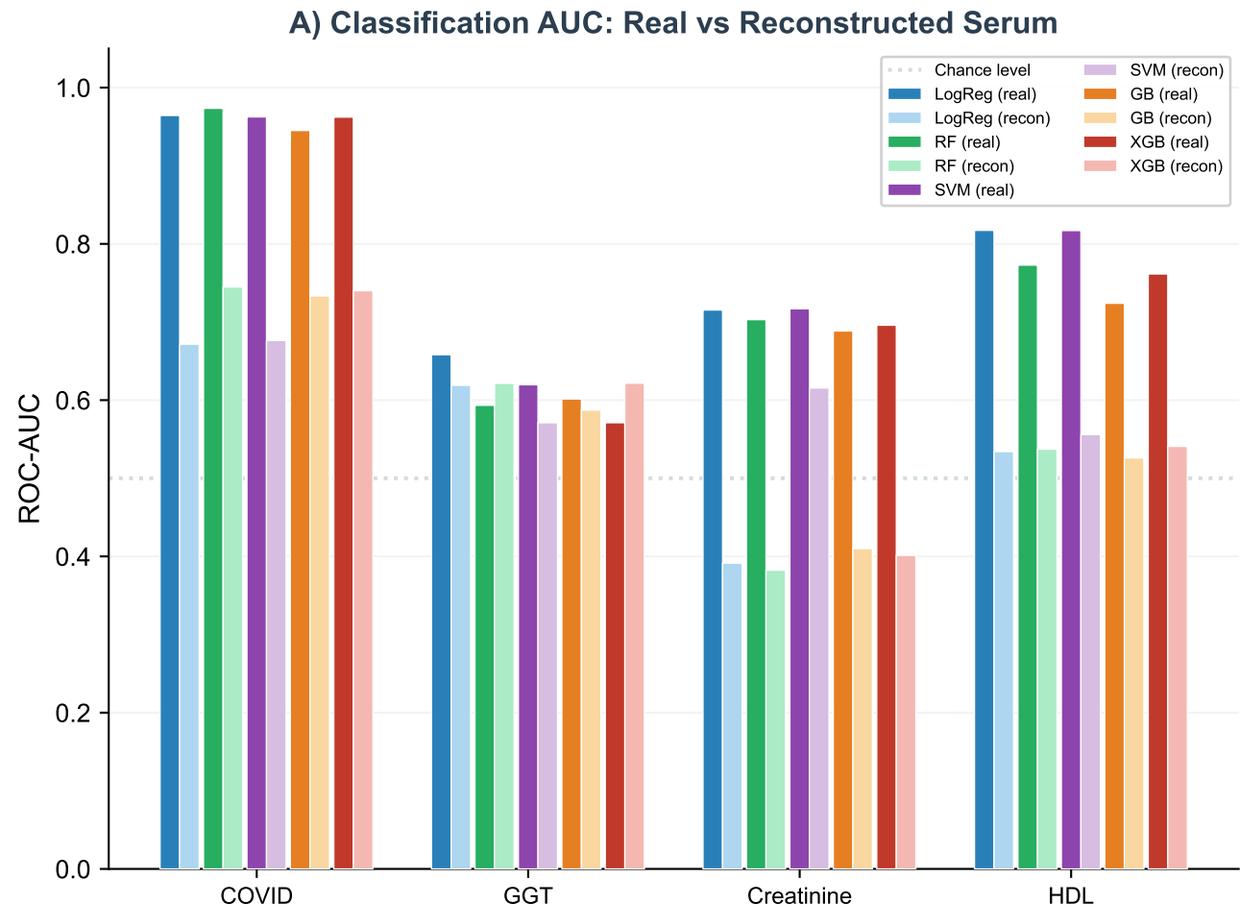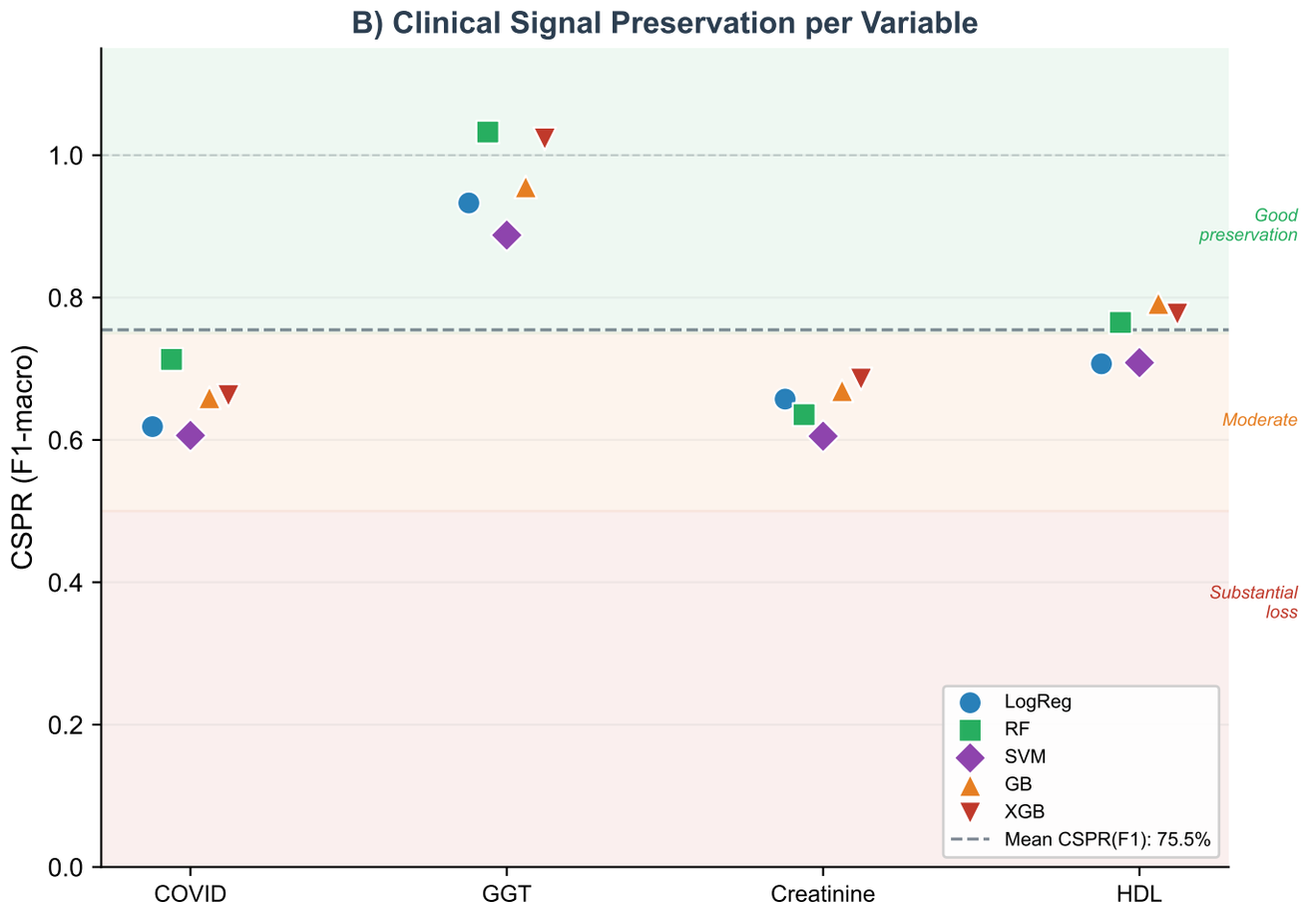Clinical Signal Preservation Analysis